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The man who stops advertising to save money is the man who saves the clock to stop time.

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-Henry Ford



closed environment. Fumes must be clean when being discharged into the atmosphere. One solution to replace the wires and solder is the use of water-soluble, electrically-conductive moulded plastics.

Thermal considerations for PCB material selection. Two factors that impact PCBs are power and heat. Therefore it is crucial to determine the thresholds for each. This can be done by assessing the thermal conductivity

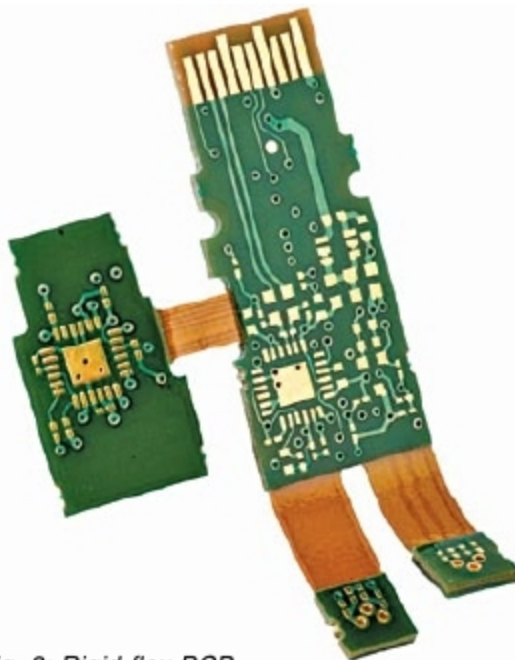


Fig. 3: Rigid-flex PCB
(Credit: www.archercircuits.com)

of the PCBs through the length of the material.

PCB material with low thermal conductivity causes heat, which can be a huge drawback for heat-intensive applications.

Selecting PCB materials

PCBs are manufactured in single- as well as double-sided varieties, some of which are copper-clad and others use aluminium for military and aerospace, automobile and medical industries applications. For these specific areas, material used should give maximum performance.

PCB materials are selected for their light weight, quality or their ability to handle high amounts of power. Since material levels correlate to performance levels, it is crucial to determine which functions need to be compared with one another when selecting PCB materials.

Majority of flex boards consist of Kapton, a polyimide film that has qualities such as heat resistance, dimensional consistency and

dielectric constant of only 3.6. Kapton comes in three Pyralux versions: FR, non-flame retardant (NFR) and adhesive-less, high performance (AP).

Quality is important in the construction of any type of board for home electronics or industrial equipment. Components such as PCBs should offer superior performance for the expected lifespan. Electronic devices, microwaves and other household devices rely on PCB technology to stay in working condition.

PCB for LED lighting. LED PCB boards generate heat while in operation. Hence, LED chips are mounted on a base made of metals like aluminium, copper or a mixture of alloy, and coated with high reflective surfaces to achieve optimum heat management and to increase light output. This keeps the heat-generating components cool and improves their ability to dissipate heat. This results in increased performance and life of the LEDs.

Consequently, metal-core PCBs (MC-PCBs) are chosen for LED applications. These include a thin layer of thermally-conductive dielectric material that can transfer and dissipate heat with much greater efficiency than traditional rigid-PCBs. FR-4 materials include a thermal aluminium-clad layer to efficiently dissipate heat. Whereas, MC-PCB materials are developed for higher power.

The future

Miniaturisation of electronic products continues to drive PCB manufacturing towards smaller and more densely-packed boards with increased electronic capabilities. Advancements include 3D moulded plastic boards and increased use of IC chips. Such advancements will keep the manufacture of PCBs a dynamic field for years to come.

Planning a well-designed layer stackup can both minimise electromagnetic radiation and stop circuits from being interfered by external noise sources. It can also reduce signal crosstalk and impedance mismatch issues. Development of modern electronics demands new PCB features such as lightweight, miniaturisation, better functionality and reliability, high speed, and longer life. **EFY**